

SINGLE-CELL SEQUENCE-TO-FUNCTION MODELING OF PANCREAS AND IMMUNE CELLS FOR LARGE-SCALE VARIANT EFFECT PREDICTION IN TYPE 1 DIABETES

Jeff Jauregui, Weston Ellison, Kyle Gaulton, Graham McVicker
Salk Institute for Biological Studies, UC San Diego

T1D risk is mostly non-coding

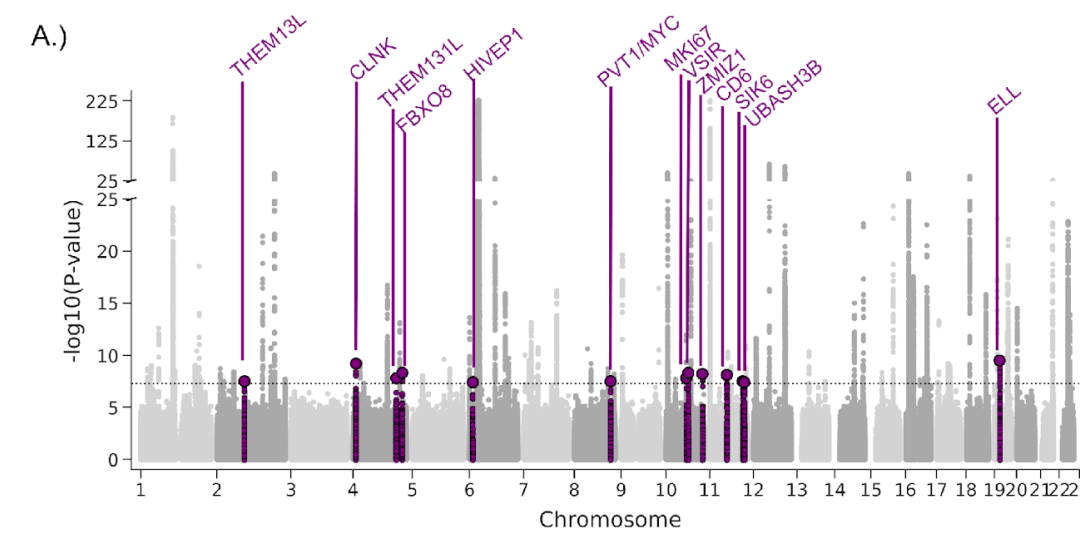


Fig 1. T1D GWAS: 165 risk signals, $N=817,718$ — mostly non-coding (McGrail 2024)¹.

Sequence models predict cell-specific regulation

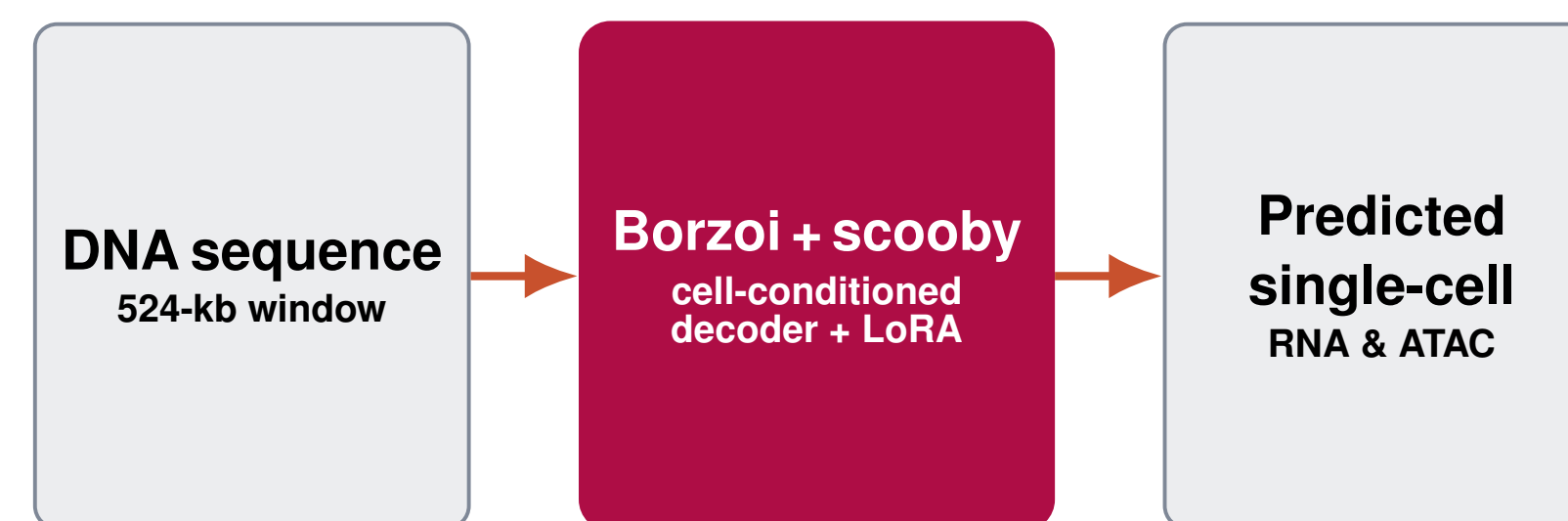


Fig 2. *Borzoi*²+*scooby*³ predict single-cell RNA/ATAC from sequence via *LoRA*.

Predicted effects match observed eQTLs

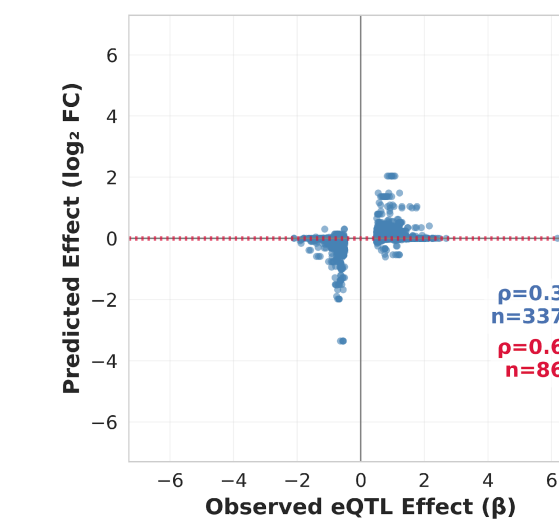


Fig 3. Predicted vs. observed eQTLs (*OneK1K4*): $\rho=0.32$ ($n=3,377$), 0.65 high-conf ($n=860$).

Rare-variant effects are cell-type-specific

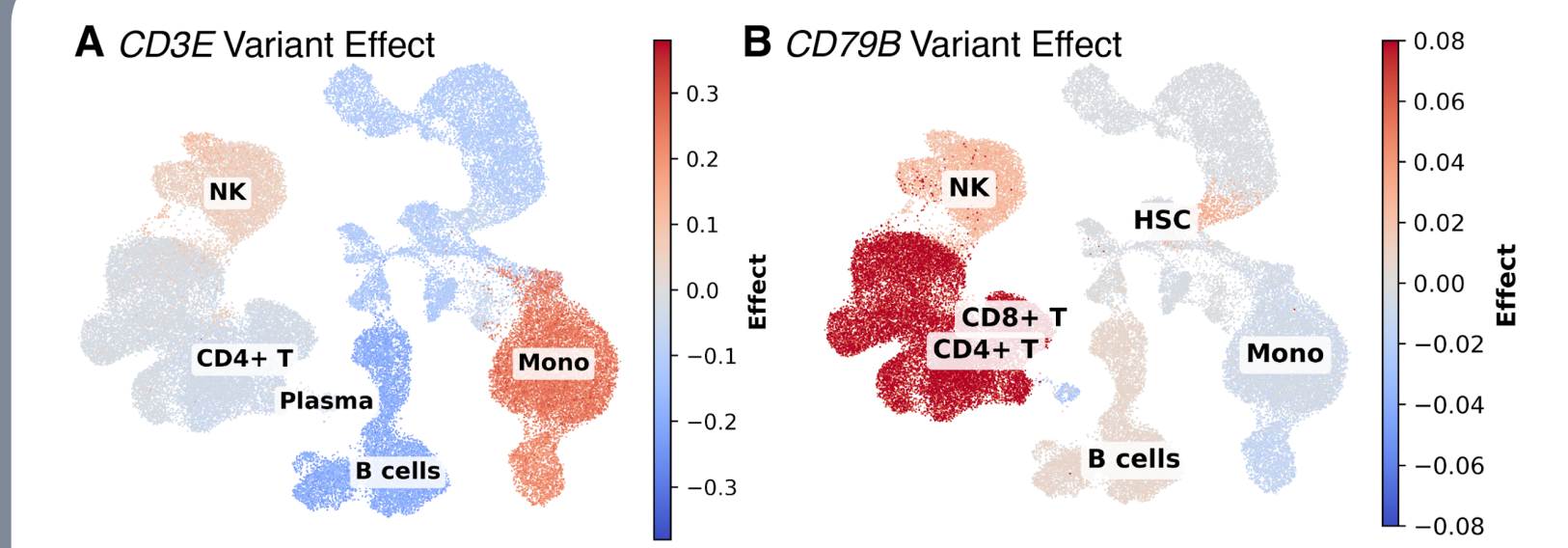


Fig 4. ~400k rare variants: *CD3E* (T/NK) vs. *CD79B* (B-cell) effects differ by cell type.

A 733k-cell islet + immune atlas

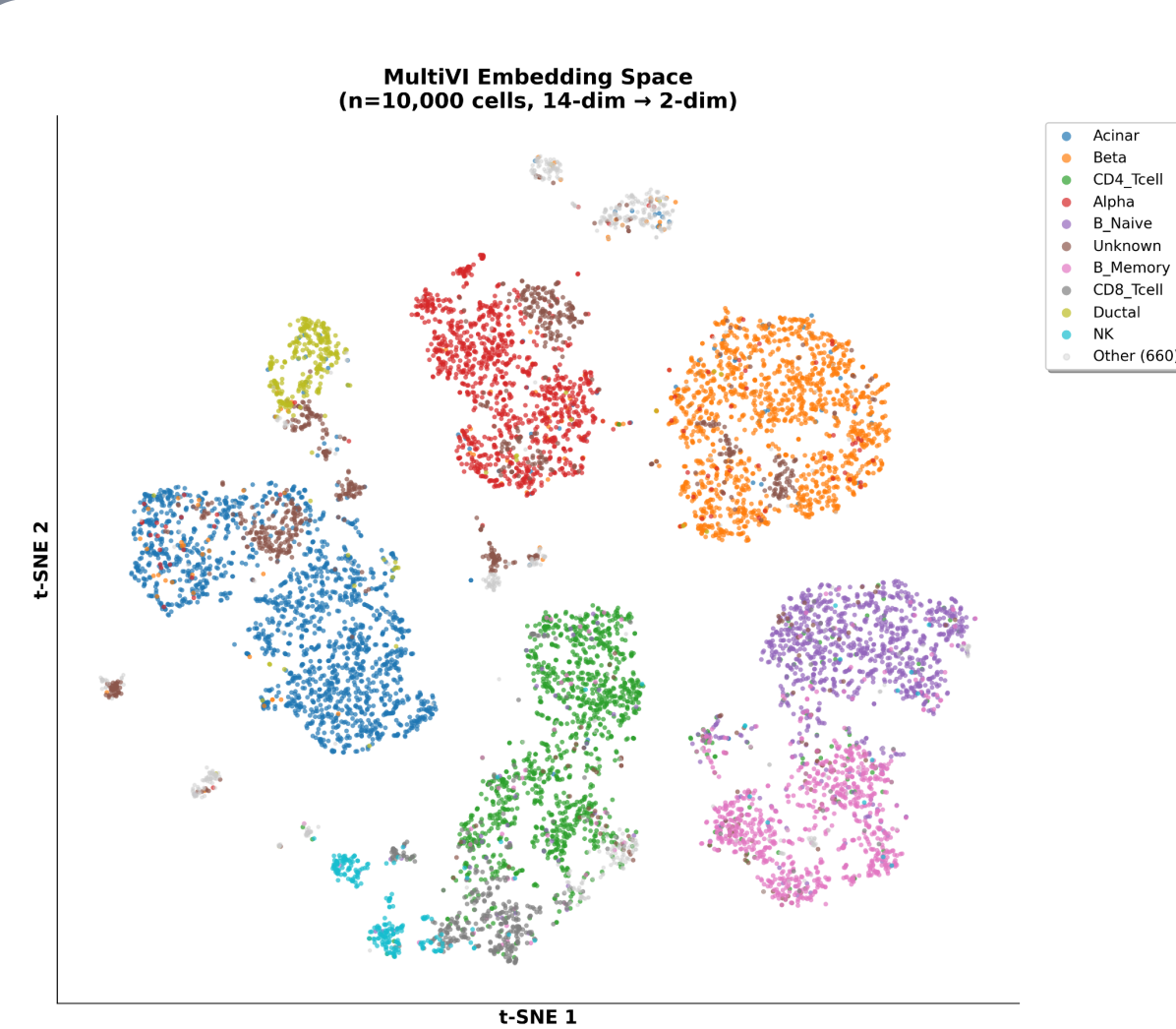


Fig 5. MultiVI⁵ atlas: 733k cells, 168 donors, 5 cohorts; islet + immune states shared across donors.

Accurate across 130 held-out genes

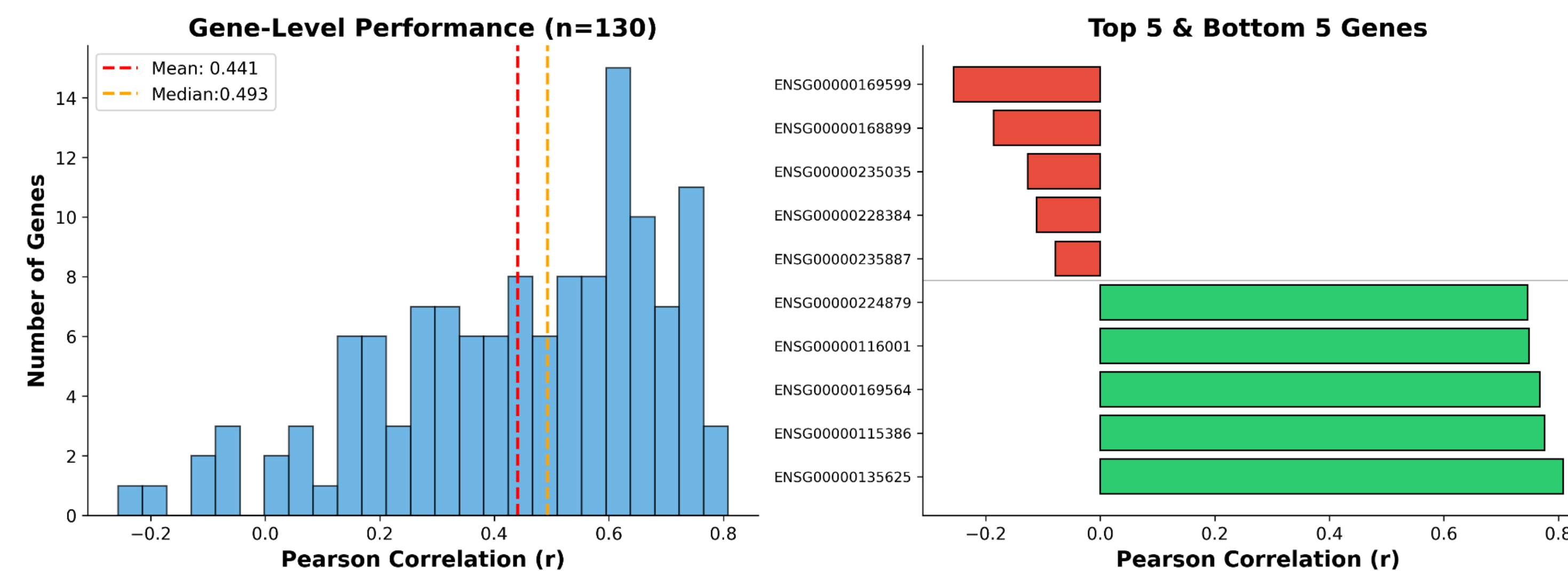


Fig 6. 130 held-out genes: mean Pearson $r=0.44$, median 0.49.

Accuracy strongest in well-represented states

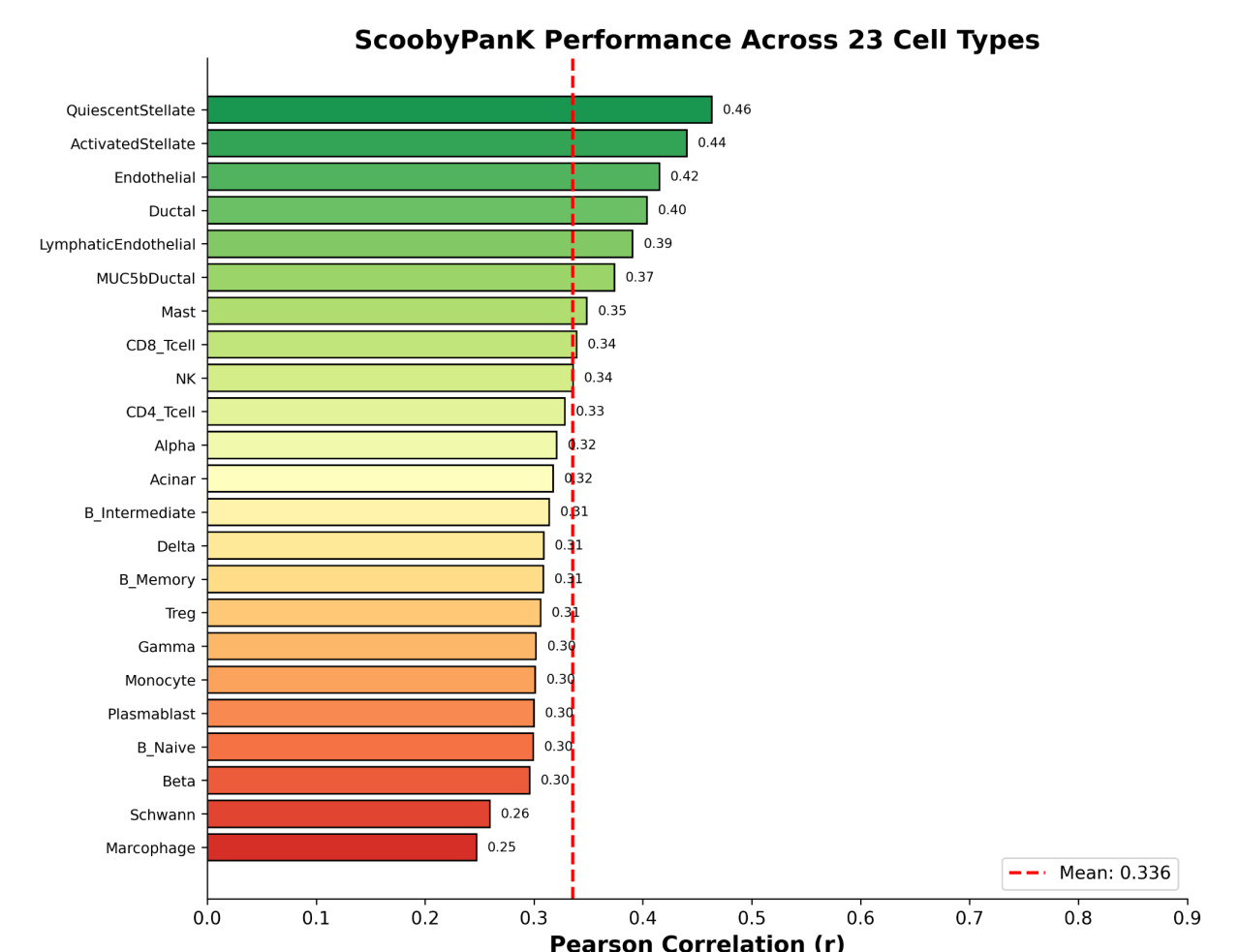
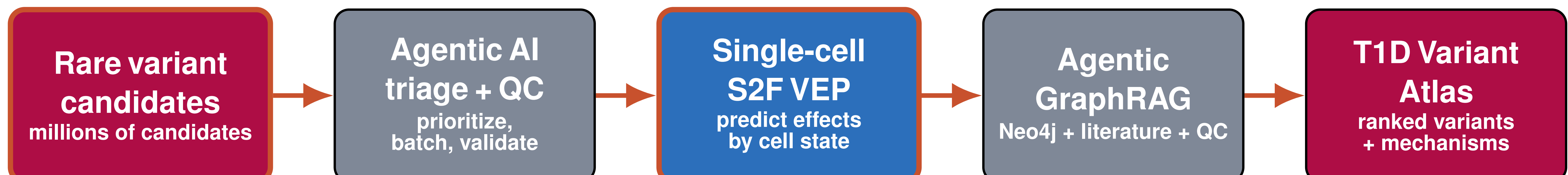


Fig 7. 23 cell types, mean $r=0.34$; highest in abundant islet stromal states.

Next: agentic AI-driven rare-variant prioritization and interpretation at scale



Links, funding & contact



Full references, methods & datasets via QR. McVicker & Gaulton Labs · jjjauregu@ucsd.edu · Funding: Salk AWS Moon Shot, Chapman, UCSD PPPF, NIH F31, STARS, Sloan; GraphRAG via Anthropic AI for Science.

UC San Diego

salk
Where cures begin.